

Supplementary Material: AGILE: Hand-Object Interaction Reconstruction from Video via Agentic Generation

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CCS Concepts: • **Computing methodologies** → **Reconstruction**; *Mesh geometry models*.

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1 Implementation Details

Pipeline and dependencies. Our framework integrates several state-of-the-art foundation models. Crucially, we utilize **Gemini 3 Pro** [Comanici et al. 2025] as the VLM agent responsible for keyframe selection and rigorous quality assessment. Guided by this agent, multi-view image synthesis is performed using the Gemini 2.5 Flash image generation model [Comanici et al. 2025], while subsequent 3D mesh and texture generation are handled by Hunyuan 3D [Team 2025a,b]. For preprocessing, geometric priors including metric depth and segmentation masks are extracted via MoGe-2 [Wang et al. 2025] and SAM2 [Ravi et al. 2024], respectively, with the initial object pose estimated by FoundationPose [Wen et al. 2024]. All differentiable

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rendering is implemented using PyTorch3D [Ravi et al. 2020]. The entire pipeline, from preprocessing to optimization, is designed to run efficiently on a single NVIDIA 4090 GPU.

Automated retopology and UV unwrapping. After the initial 3D mesh is generated by Hunyuan 3D, we apply automated retopology and UV unwrapping to produce a clean, simulation-ready asset. Specifically, we leverage Blender's built-in quad-remeshing algorithm to convert the raw triangular mesh into a uniform quad-dominant topology with a target face count of approximately 10,000 faces. This step ensures the mesh has a regular, artifact-free topology suitable for physics simulation and collision detection. Subsequently, we perform automatic UV projection using Blender's Smart UV Project, which segments the surface into non-overlapping UV islands and minimizes distortion. The original texture is then baked onto the new UV layout via ray-cast projection, preserving the high-frequency surface details from the generative model. This fully automated process eliminates the need for any manual post-processing.

Online optimization strategy. Our method optimizes each frame sequentially via a two-stage alternating approach to ensure stability. Stage one optimizes hand translation (up to 200 iterations), followed by object 6D pose refinement (up to 400 iterations) with fixed hand parameters. exclusively for the onset frame, we optimize object scale by minimizing $\mathcal{L}_{\text{mask}}$ and the ray-traced contact loss $\mathcal{L}_{\text{contact}}$. During scale optimization, object rotation is fixed, and translation is constrained to the visual ray passing through the object center, thereby distinctively regularizing the solution space. Convergence is reached when the moving average of parameter variations, including scale when active, is less than 10^{-4} .

Optimizers and learning-rate schedule. For optimization, we employ separate Adam optimizers [Kingma 2014] for rotation and translation parameters. The object optimization uses fixed learning rates of 2×10^{-3} for rotation and 1×10^{-3} for translation. Similarly, hand optimization uses fixed learning rates of 1×10^{-3} for translation. We adopt constant learning rates throughout optimization, as we empirically observe that learning-rate decay does not lead to

measurable accuracy improvements in our setting, while increasing the number of iterations required for convergence.

Loss terms and weights. The loss function employs the following weights: $\mathcal{L}_{\text{mask}}$ (L2) is weighted at 5.0, $\mathcal{L}_{\text{dino}}$ (L1) at 10.0. For physical interaction modeling, the ray-traced contact loss $\mathcal{L}_{\text{contact}}$ receives a weight of 5.0. Additionally, the relative position interaction loss $\mathcal{L}_{\text{interact}}$ is weighted at 400 for HO3D dataset and 200 for DexYCB dataset, with a maximum distance threshold of 0.05m to prevent spurious long-range attractions between hand and object. For the SDF-based distance-weight gating function $w_i = 1 - \tanh(\sigma \cdot d_i)$ defined in the main paper, we set $\sigma = 40$.

Interaction Onset Frame selection. For HO3D, we identify the interaction onset as the earliest frame where the object exhibits significant motion while remaining fully visible. We quantify the motion by measuring the variation of the non-occluded object mask between consecutive frames:

$$r_i = \frac{\|(\mathcal{M}_o^{i+1} - \mathcal{M}_o^i) \odot (1 - \mathcal{M}_h^{i+1}) \odot (1 - \mathcal{M}_h^i)\|_1}{\|\mathcal{M}_o^i\|_1 + \varepsilon}, \quad (1)$$

where $\mathcal{M}_o$ and $\mathcal{M}_h$ denote the object and hand masks, respectively. The numerator captures the change in visible object pixels, while the denominator normalizes by the object area ($\varepsilon = 10^{-8}$ ensures stability). We select the first frame i that satisfies two criteria: (1) the motion ratio r_i exceeds a threshold $\tau = 0.025$, and (2) the object mask does not touch the image boundaries. The latter ensures the object is not truncated, thereby guaranteeing reliable geometric supervision.

2 Why Start at Interaction Onset?

We choose the interaction onset frame as the optimization anchor for two key reasons. First, metric scale alignment: since monocular object reconstruction suffers from scale ambiguity, the physical contact allows us to leverage the hand’s reliable metric scale to constrain and propagate the correct object size. Second, loss efficacy: the contact-based losses ($\mathcal{L}_{\text{contact}}$) are physically meaningful only when interaction occurs. Initiating optimization at this frame ensures these geometric constraints provide valid gradients for precise hand-object alignment.

3 Clarification on the Static Camera Assumption

We clarify that the static camera assumption in our framework is strictly limited to the automatic selection of the Interaction Onset Frame (IOF), rather than the core reconstruction pipeline itself.

Why a static camera for IOF detection? As described in Sec. 1, we identify the IOF by measuring the 2D pixel displacement of the unoccluded object mask between consecutive frames. This specific heuristic assumes a fixed camera to reliably distinguish genuine object movement from camera ego-motion. Under a moving camera, apparent mask displacement could arise from viewpoint change rather than object manipulation, leading to incorrect IOF identification.

Core pipeline independence. Crucially, our core pipeline—Agentic Generation (Sec. 3.1), Pose Initialization (Sec. 3.2), and Contact-Aware Tracking (Sec. 3.3)—operates entirely in the camera coordinate system. Since all optimization variables (hand pose, object pose, and interaction constraints) are defined relative to the camera frame, camera ego-motion is mathematically irrelevant to our optimization. This means the reconstruction quality is not affected by whether the camera is static or dynamic, as long as a valid IOF is identified.

Extension to dynamic cameras. To handle dynamic cameras (e.g., first-person or egocentric videos), one simply needs to substitute the IOF selection step with alternative methods—such as VLM-based action recognition (e.g., detecting the moment of grasping), off-the-shelf contact predictors, or manual specification. Camera intrinsics and extrinsics for each frame can be recovered using existing structure-from-motion or SLAM pipelines (e.g., MegaSaM [Li et al. 2025]). The rest of our framework requires absolutely zero modification. We view this modularity as a strength: the IOF selection is a replaceable preprocessing component, while the core method is camera-motion agnostic.

4 Details of Fine-Grained Ablation Variants

The fine-grained ablation results on Agentic Multi-view Generation are reported in Table 3 of the main text (variants b_1 , b_2). Here we describe the ablation configurations in detail:

(1) *Heuristic Selection* (b_1): We replace the VLM-based keyframe selection with a heuristic that simply chooses the frames with the largest object mask areas. Multi-view synthesis and all subsequent steps remain unchanged. This isolates the contribution of the VLM’s spatial reasoning for selecting geometrically informative views.

(2) *w/o Multi-View Synthesis* (b_2): We retain VLM keyframe selection but skip the multi-view synthesis stage, directly feeding the single selected keyframe to the 3D generation model. This isolates the contribution of multi-view consistency for faithful 3D reconstruction.

Both variants use the *exact same* foundation models (Gemini, Hunyuan3D) as the full pipeline to ensure a fair comparison.

5 Sequence Used

We evaluate our method on sequences from the DexYCB [Chao et al. 2021] and HO3D [Hampali et al. 2020] datasets. Specifically, as shown in Table 1 and Table 2, we randomly select 18 sequences from HO3D and 20 sequences from DexYCB, covering a diverse range of object types and hand-object interaction patterns. For the HO3D dataset, we begin processing from the identified IOF to ensure meaningful interaction modeling. HO3D sequences vary in length from 800 to 1600 frames; for computational efficiency, we cap the processing at 1024 frames per sequence. For the DexYCB dataset, each selected sequence spans approximately 80 frames. Unless otherwise specified, we uniformly subsample all sequences with a stride of 5 (i.e., processing every fifth frame), which balances temporal coverage with computational cost. These sequences are chosen to provide comprehensive coverage of common manipulation scenarios and object categories, enabling robust evaluation of our method’s generalization capabilities across different interaction types and geometric configurations.

Table 1. Sequences selected from the HO3D datasets.

Sequence ID	Object	IOF Index
ABF12	bleach cleanser	100
ABF14	bleach cleanser	100
BB12	banana	200
BB13	banana	130
GPMF12	potted meat	105
GPMF14	potted meat	100
GSF13	scissors	100
GSF14	scissors	80
MC1	cracker box	0
MC4	cracker box	0
MDF12	power drill	150
MDF14	power drill	250
ShSu10	sugar box	50
ShSu12	sugar box	110
SM2	mustard	0
SM4	mustard	0
SMu1	mug	20
SMu40	mug	0

Table 2. Sequences selected from the DexYCB datasets.

Sequence ID	Object
20200709-subject-01/20200709_141754/836212060125	master chef can
20200709-subject-01/20200709_142853/836212060125	master chef can
20200813-subject-02/20200813_154204/836212060125	bleach cleanser
20200813-subject-02/20200813_153453/836212060125	power drill
20200820-subject-03/20200820_141856/836212060125	potted meat
20200820-subject-03/20200820_143330/836212060125	mug
20200903-subject-04/20200903_103554/836212060125	master chef can
20200903-subject-04/20200903_113012/836212060125	wood block
20200908-subject-05/20200908_144138/836212060125	tomato soup can
20200908-subject-05/20200908_152416/836212060125	large clamp
20200918-subject-06/20200918_113441/836212060125	sugar box
20200918-subject-06/20200918_114747/836212060125	potted meat
20200928-subject-07/20200928_153800/836212060125	pitcher base
20200928-subject-07/20200928_144300/836212060125	mustard bottle
20201002-subject-08/20201002_110940/836212060125	pitcher base
20201002-subject-08/20201002_105558/836212060125	mustard bottle
20201015-subject-09/20201015_143700/836212060125	mustard bottle
20201015-subject-09/20201015_142940/836212060125	cracker box
20201022-subject-10/20201022_111745/836212060125	mustard bottle
20201022-subject-10/20201022_111209/836212060125	sugar box

6 Analysis on Results of Baselines

On the DexYCB dataset, both baselines exhibited varying degrees of failure. Specifically, HOLD encountered issues in some sequences where it failed to obtain the hand/object mesh due to inaccurate poses, which prevented the geometric structure from being effectively learned. Meanwhile, MagicHOI was unable to complete colmap reconstruction in most sequences and thus could not obtain stable poses for subsequent processing. On the other hand, MagicHOI’s hand-object alignment relies on the geometry generated by its generative model. Due to the limitations of its capabilities, it fails

to produce reasonable geometric structures in some challenging cases, leading to the failure of the final results.

7 Impact of texture refinement on pose initialization.

As shown in Table 3, texture quality plays a decisive role in the performance of the subsequent pose estimation. FoundationPose [Wen et al. 2024] adopts an analysis-by-synthesis approach, estimating the 6D pose by comparing the similarity between the input image and renderings of the object mesh. Consequently, accurate appearance modeling is critical for minimizing the domain gap between synthesized and observed views. Relying solely on the initial coarse texture leads to significant appearance discrepancies, which confuse the scoring network and result in a high mean rotation error of 118.41° . In contrast, our texture refinement recovers high-fidelity surface details that closely match the visual observation. This photorealism enables FoundationPose to compute more reliable similarity scores, drastically reducing the rotation error to 81.55° and improving the ADDS accuracy from 18.52% to 27.42%. These results validate that precise texture recovery is not merely a visual enhancement but a prerequisite for robust geometry-based pose estimation.

8 Comparison with Generative 3D Initialization

Table 4 presents a comparative analysis against SAM3D [Team et al. 2025], a state-of-the-art method that jointly estimates shape and pose from a single image.

Experimental Setup. Given that SAM3D operates on a single-frame basis, we evaluate it on every 5th frame across all 18 scenes in the HO3D dataset. Conversely, our method generates a single canonical mesh using VLM-guided multi-view synthesis and estimates the pose for each frame using FoundationPose initialized with MoGe-2 metric depth.

Analysis. The fundamental limitation of SAM3D lies in *geometric hallucination* due to occlusion. As a single-view model, it is forced to reconstruct unobserved regions based on learned priors rather than temporal evidence. This often leads to geometries that are topologically plausible but inconsistent with the ground truth, as evidenced by the high Chamfer Distance (1.35 vs. 0.27 cm^2). Such geometric inconsistency inherently compromises pose estimation accuracy, as the predicted pose lacks a faithful 3D reference to align with the input image, resulting in a large rotation error of 118.71° . In contrast, by aggregating multi-view cues, our approach maintains rigorous geometric consistency, enabling reliable, metric-scale pose estimation even under heavy occlusion. A qualitative comparison of the reconstructed geometry is illustrated in Figure 1.

9 Computation Cost

Our optimization process is computationally efficient. On a single NVIDIA RTX 4090 GPU, each frame requires approximately 30–50 seconds for optimization. Unless otherwise specified, we process every fifth frame of each sequence. As a result, the total computation time scales linearly with the sequence length. On average, a sequence of 1000 frames (200 frames after downsampling) can be processed in approximately 1.5 hours, which is consistent with the overall runtime reported in Table 5.

Table 3. **Impact of Texture Refinement on FoundationPose Initialization.** We compare the pose estimation performance on the HO3D dataset using the original coarse texture (w/o Refinement) versus our refined texture (w/ Refinement). FoundationPose yields significantly lower rotation errors and higher accuracy scores when supplied with the refined, photorealistic texture.

Method	R Err. (°)↓	T Err. (mm)↓	ADD (%)↑	ADDS (%)↑	MSSD↑	MSPD↑	VSD↑	AR↑
w/o Refinement (Coarse Texture)	118.41	10.37	0.42	18.52	5.70	2.93	2.86	3.83
w/ Refinement (Ours)	81.55	10.35	6.36	27.42	19.09	7.25	5.28	10.54

Table 4. **Comparison with Generative 3D Baseline (SAM3D).** Quantitative comparison of pose initialization and geometric fidelity against the single-view baseline SAM3D. Our initialization approach significantly reduces geometric error and improves pose alignment across all metrics.

Method	R Err. ° ↓	T Err. (mm)↓	CD (cm ²)↓	F@5 (%)↑	F@10 (%)↑	ADDS (%)↑	MSSD↑	MSPD↑	VSD↑	AR↑
SAM3D	118.71	24.85	1.35	67.01	87.44	9.06	0.80	0.00	5.62	2.47
Our Initialization	81.55	10.35	0.27	86.63	97.77	27.42	19.09	7.25	5.28	10.54

Table 5. Comparison of computational cost for a 200-frame sequence on a single RTX 4090 GPU. Our method requires significantly less preprocessing and total runtime than existing methods.

Method	Preprocessing	Optimization	Total
HOLD	~65 min	~24 h	~25.1 h
MagicHOI	~25 min	~2 h	~2.4 h
Ours	~21–29 min	~1.75 h	~2.1–2.2 h



Fig. 1. Qualitative comparison with SAM3D [Team et al. 2025]. Our method significantly outperforms SAM3D in terms of both geometry and texture.

Detailed runtime breakdown. We report the end-to-end runtime for a representative 200-frame sequence (1000 raw frames at stride 5)

on a single NVIDIA RTX 4090 GPU. Despite introducing a VLM critic in the generation loop, our total pipeline time remains competitive with or faster than existing baselines:

- **HOLD:** ~25.1 hours (Preprocessing: ~65 min; Optimization: ~24 h)
- **MagicHOI:** ~2.4 hours (Preprocessing: ~25 min; Optimization: ~2 h)
- **Ours:** ~2.1–2.2 hours (Preprocessing: ~21–29 min; Optimization: ~1.75 h)

Breakdown of our preprocessing (~21–29 min): The Agentic Generation stage accounts for ~16–24 minutes, depending on the number of rejection sampling iterations ($n \in [1, 3]$). This includes keyframe selection (~2 min), 3D generation (~10 min), multi-view synthesis (~2–6 min), and texture refinement (~2–6 min). The remaining preprocessing takes ~5 minutes: mask extraction via SAM2 (~20 s), metric depth via MoGe-2 (~88 s), hand pose via WiLoR (~24 s), and FoundationPose initialization (~160 s).

10 Details of VLM-Guided 3D Generation

In this section, we provide the detailed prompt specifications and evaluation protocols used in our agentic generation pipeline. Our framework leverages a Vision-Language Model (VLM) as an intelligent supervisor to guide three critical stages: (1) informative keyframe selection, (2) consistent multi-view synthesis, and (3) high-fidelity texture refinement. For each stage, we present the system instructions, task definitions, and representative VLM responses. The prompts presented below are the exact instructions used in our pipeline to ensure reproducibility.

10.1 Stage 1: Informative Keyframe Selection

The objective of this stage is to filter the input video stream and identify a minimal set of frames that maximize geometric coverage while minimizing occlusion.

10.1.1 Prompt Specification.

{

```
"system_role": "You are an expert in computer
vision, specializing in selecting frames that
best showcase the complete appearance of an
object from multiple images.",
"task": "Select {num_frames} images from the
following candidate frames. The selected images
must:
1. Clearly display different angles of the object.
2. Maximize coverage of the object's complete
appearance (front, back, left, right, top,
bottom).
3. Be sharp, with the object fully visible and
minimal occlusion.
4. Have the maximum possible viewpoint difference
between selected frames.
5. Feature the object occupying a relatively large
portion of the frame.",
"response_format": "
{
  "selected_frames": [1, 5, 10, 15], // Indices of
selected frames
  "reasoning": "Reason for selection",
  "coverage": {
    "front": true,
    "back": true,
    "left": true,
    "right": false
  }
}",
"instruction": "Please reply strictly in JSON
format without adding other explanatory text."
}
```

Listing 1. Prompt for Keyframe Selection

10.1.2 Example Response.

```
{
  "selected_frames": [1, 5, 13, 21],
  "reasoning": "Frame 1 is selected as the standard
front view, clearly displaying the main brand
label. Frame 5 shows the back of the object,
clearly presenting the ingredients list and
nutritional information. Frame 13 provides the
side profile, effectively showcasing the object
's depth, thickness, and side curves. Frame 21
captures the bottom perspective, supplementing
this often-overlooked angle. All four images
are high-definition, have a large object-to-
frame ratio, and exhibit significant viewpoint
differences, achieving maximum multi-angle
coverage.",
  "coverage": {
    "front": true,
    "back": true,
    "left": true,
```

```
"right": false,
"top": false,
"bottom": true
}
}
```

Listing 2. VLM Response for Keyframe Selection

10.2 Stage 2: Multi-View Synthesis and Validation

Based on the selected keyframes, we generate four orthogonal views. To ensure consistency, a VLM critic evaluates the generated views against strict geometric and texture criteria.

10.2.1 Generation Prompts.

```
{
  "multiview_generation": {
    "static_object": "Generate a complete image of the
water bottle in the picture from four
perspectives: front, back, left, and right.
Require photorealism. Present as a four-view
grid. Remove the plate and extraneous objects.
Pure white background. The image must not
contain text indicating viewpoint direction (e.
g., front, back, left, right).",
    "hand_held_object": "Generate a complete image of
the object held in the hand from four
perspectives: front, back, left, and right.
Require photorealism. Present as a four-view
grid. Remove the hand and hallucinate/complete
the regions occluded by the hand. Pure white
background. The image must not contain text
indicating viewpoint direction. The object size
should be consistent across all four views."
  }
}
```

Listing 3. Prompts for Multi-View Generation

10.2.2 Validation Protocol (The "Critic").

```
{
  "validation": {
    "system_role": "You are an expert in 3D modeling,
material analysis, and multi-view image quality
assessment. Please carefully analyze the
following images:",
    "task_description": "Task:
1. The first image(s) are the original input,
showing the appearance, texture, and material
of an object.
2. The last image is the generated 'four-view'
image, which should display the complete image
of the object from four different perspectives
(front, back, left, right) while preserving the
original visual attributes.",
```

```

"evaluation_criteria": "Criteria:

Level 1: Veto Items
1. Text Check: Does the generated image contain
   any text, labels, or viewpoint descriptions (e.
   g., front, back)? If yes, terminate evaluation;
   result is invalid.

Level 2: Core Dimension Scoring (0-10)
2. Geometry & View Correctness (Weight: 30%): Are
   viewpoints correct? Is orientation consistent (
   no rotation)? Any rotation results in large
   deductions.
3. Texture & Material Fidelity (Weight: 20%): Are
   surface textures (e.g., patterns) and material
   properties (e.g., reflection) consistent with
   the original?
4. Geometric Detail Integrity (Weight: 20%): Are
   key geometric details (chamfers, holes,
   embossing) preserved?
5. Feature Consistency (Weight: 15%): Is it the
   same object in terms of shape, style, and color
   ?
6. Image Quality (Weight: 15%): Is the image clear
   , noise-free, and on a pure white background?

Level 3: Deductions
- Rotated views: -3 points each.
- Poor layout: -1 to -2 points.",
"response_format": "JSON format containing:
  is_valid, score_overall, score_breakdown,
  has_text, rotated_views,
  improvement_suggestions, summary_feedback, etc.
",
"instruction": "Reply strictly in JSON. First
  check veto items."
}
}

```

Listing 4. Prompt for Multi-View Validation

10.3 Stage 3: Texture Refinement and Agentic Editing

After lifting the multi-view images to 3D and unwrapping UVs, we perform a texture refinement step. A VLM critic detects artifacts (e.g., hallucinated content, missing details) and guides a subsequent editing pass.

10.3.1 Texture Validation Protocol.

```

{
  "texture_validate": {
    "system_role": "You are an expert in 3D texture
      mapping analysis, material comparison, and
      image consistency assessment. Carefully analyze
      the images.",
    "task_description": "Task:

```

```

1. Image 1: The four-view image.
2. Image 2: The generated Texture Map.
3. Evaluate if the texture map accurately
   reproduces all texture information from the
   four-view image, focusing on completeness,
   fictional content, and correspondence.",
"evaluation_criteria": "Criteria:

```

```

Level 1: Veto Items
- Invalid texture map (blank, pure color, severe
  distortion).
- Key features completely missing.

Level 2: Core Dimensions
1. Completeness (Weight: 30%): Are all visible
  textures present? (Deduction: -2 per major
  missing item).
2. Accuracy/No Fiction (Weight: 25%): Does it
  contain hallucinated content not present in the
  source? (Severe penalty: -3 to -5 points).
3. Correspondence (Weight: 20%): Are textures
  mapped to correct UV islands?
4. Color/Material (Weight: 15%): Consistency in
  color, shading, and saturation.
5. Fidelity (Weight: 10%): Resolution and detail
  preservation.

Level 3: Extra Deductions
- Seams, repetitions, stretching.",
"response_format": "JSON format containing:
  is_valid, score_overall,
  fictional_content_analysis,
  missing_content_analysis,
  texture_coverage_analysis, etc.",
"instruction": "Strictly follow JSON format. Pay
  special attention to 'fictional content', any
  texture appearing in the map but not in the
  source views is a severe error!"
}
}

```

Listing 5. Prompt for Texture Map Validation

10.3.2 Example Validation Result (Invalid Case Detected).

```

{
  "is_valid": false,
  "score_overall": 0.0,
  "fictional_content_analysis": {
    "has_fictional_content": true,
    "fictional_items": [
      {
        "type": "severe",

```

```

"description": "The entire back texture is a
fabricated copy of the front label instead of
the nutrition facts panel shown in the source.",

"location_in_texture": "The second large body
island in the texture map.",
"deduction": 5
}
],
"total_fictional_deduction": 5
},
"summary_feedback": "Invalid. Critical error: the
back side texture is a fabrication (copy of the
front). This makes the model unusable for pose
estimation from rear viewpoints."
}

```

Listing 6. Example Validation Result detecting hallucinations

10.3.3 Refinement Prompt Construction.

Based on the negative feedback from the validation step, we programmatically construct a structured refinement command to guide the image-to-image editing model.

```

{
"texture_refinement_task": {
"base_instruction": "Below are object images from
multiple views and their texture map. Please
edit the texture map based on the following
critique:",
"injected_feedback": "Invalid. Critical error: the
back side texture is a fabrication (copy of
the front). This makes the model unusable for
pose estimation from rear viewpoints.",
"geometric_constraints": "Do not change the layout
or shape of UV islands. Only align the content
with reference images. The bottom-left area
corresponds to the bottom view.",
"specific_corrections": "Change the purple/black
artifact regions to match the red color of
other parts. Adjust tone for vibrancy. Fix text
and patterns."
}
}

```

Listing 7. Structured Input for Texture Refinement

References

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